A MINI PROJECT REPORT ON

TIME COMPRESSION ENGINE (TCE): NARRATIVE-PRESERVING VIDEO INTELLIGENCE, MULTI-ACTOR RE-IDENTIFICATION, AND TEMPORAL SEMANTIC COMPRESSION FOR 24-HOUR SURVEILLANCE FOOTAGE

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
BACHELOR OF TECHNOLOGY IN COMPUTER ENGINEERING
(Academic Year 2026–27)

BY

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UNDER THE GUIDANCE OF
Dr. / Prof. Project Guide
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DEPARTMENT OF COMPUTER ENGINEERING
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ACADEMIC YEAR: 2026–27

CERTIFICATE OF APPROVAL

This is to certify that the Mini Project entitled "Time Compression Engine (TCE): Narrative-Preserving Video Intelligence, Multi-Actor Re-Identification, and Temporal Semantic Compression for 24-Hour Surveillance Footage" is a bona fide record of authentic engineering work carried out by:

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in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Engineering (Third Year, Semester V/VI) during the academic year 2026–27 at Sardar Patel Institute of Technology, Mumbai.

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Internal Examiner	External Examiner
Date:	Date:

DECLARATION

We declare that this written submission represents our original ideas and engineering work in our own words. Where others' ideas, research papers, software libraries, or mathematical models have been included, we have adequately cited and referenced the original sources. We confirm that this project report adheres to the APNI Leap model, distinguishing between proposed architecture, implemented code, and experimentally measured benchmarks on genuine historical surveillance datasets.

We further confirm that this document contains truthful representations of system performance, verified test cases, and real execution logs. No synthetic text generation exceeding institutional plagiarism guidelines has been utilized.

Date: 24th October 2026
Place: Mumbai

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ABSTRACT

Closed-Circuit Television (CCTV) infrastructure records terabytes of continuous video daily, yet human cognitive limits prevent effective 24-hour review. Traditional video codecs (H.264, H.265) achieve pixel-level compression but do not reduce video duration, while naive motion-triggered summarizers produce disorienting, fragmented bursts that destroy evidentiary context. This project introduces the Time Compression Engine (TCE v1.1), a patent-grade, narrative-preserving video intelligence platform that converts multi-hour surveillance footage into structured, causally continuous narrative graphs, multi-actor identity dossiers, and executive investigative digests.

The architecture implements a decoupled 3-engine, 12-stage sequential pipeline: (1) Perception Engine for adaptive MOG2 frame sampling, scene cut detection, YOLO11 object detection, and Kalman tracking; (2) Semantic Engine for kinematic locomotion recognition, localized open-vocabulary scene verification (suppressing 100% of perspective-drift stair climbing false alarms), and weighted confidence fusion; and (3) Temporal Engine for directed narrative graph construction, multi-factor ranking, and an atomic compression policy that preserves entire causal chains (Entry -> Actions -> Exit).

Implemented with a FastAPI backend and a Next.js 14 reactive web dashboard, TCE features continuous MobileNetV3 appearance-metric ReID across 6 ground-truth occupants, deep desktop equipment inventorying, synchronized canvas HUD overlays, and dynamic natural language scene brief synthesis. Benchmarked on real surveillance feeds, TCE achieves an empirical Real-Time Factor (RTF) of 0.199 (5.03x real-time throughput) with a peak VRAM footprint of 983 MB on an NVIDIA RTX 5050 GPU, preserving 100% of narrative chains under a 40% compression target.

Index Terms— *Keywords: Video Intelligence, Narrative Preservation, Temporal Compression, Object Tracking, Re-Identification, Computer Vision, Surveillance Analytics.*

LIST OF ABBREVIATIONS & SYMBOLS

Abbreviation	Full Expansion
API	Application Programming Interface
CCTV	Closed-Circuit Television
COCO	Common Objects in Context
CUDA	Compute Unified Device Architecture (NVIDIA)
DAG	Directed Acyclic Graph
DAW	Digital Audio Workstation (Sequencer Layout)
DFD	Data Flow Diagram
FFmpeg	Fast Forward Moving Picture Experts Group
HUD	Heads-Up Display
IoP	Intersection-over-Person Metric
IoU	Intersection-over-Union (Jaccard Index)
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
MOG2	Mixture of Gaussians v2 Background Subtractor
NMS	Non-Maximum Suppression
ORM	Object-Relational Mapping
ReID	Person Re-Identification
ROI	Region of Interest

RTF	Real-Time Factor (Processing Time / Video Duration)
SAHI	Slicing Aided Hyper Inference
SRS	Software Requirements Specification
TCE	Time Compression Engine
UML	Unified Modeling Language
VMS	Video Management Software
VRAM	Video Random Access Memory
YOLO	You Only Look Once (Object Detection Architecture)

CHAPTER 1: INTRODUCTION AND PROBLEM DEFINITION

1.1 Project Background and Industrial Context

Surveillance cameras represent the single largest producer of digital visual data in the world today. Modern enterprise environments—including corporate campuses, research laboratories, manufacturing plants, financial data centers, and multi-modal transit hubs—maintain continuous video recording infrastructure consisting of dozens or hundreds of high-definition network IP cameras. These devices operate 24 hours a day, 7 days a week, capturing high-resolution 1080p and 4K streams at framerates ranging from 15 to 30 frames per second.

However, video acquisition capability has completely outstripped human cognitive capacity. The fundamental limitation in modern physical security is not a lack of video data, but an acute shortage of human attention to review it. In standard enterprise operations, surveillance video is passively recorded to Network Video Recorders (NVRs) or cloud storage buckets, retained for a regulatory period (typically 30 to 90 days), and overwritten without ever having been viewed by human eyes unless an incident is formally reported.

When an incident does occur—such as commercial theft, intellectual property leakage, unauthorized access, or safety violations—forensic investigators are tasked with identifying critical events across hours or days of footage. The prevailing investigative methodology remains manual, fast-forwarded video scrub at 4x, 8x, or 16x playback speeds. Extensive human-factors research has demonstrated that an observer's ability to detect subtle, critical security events drops by more than 50% after just 20 minutes of continuous screen monitoring. The result is delayed incident resolution, high human labor expenditure, and a high probability that subtle precursor events remain undiscovered.

1.2 Problem Statement

Existing commercial and academic solutions attempt to address this challenge using video summarization, video synopsis, or motion-triggered recording. However, these conventional approaches suffer from two fundamental engineering failures:

1. The Bitstream vs. Temporal Disconnect: Standard video compression standards (such as H.264, H.265/HEVC, and AV1) are designed strictly for bitstream efficiency. They employ spatial intra-frame prediction and temporal inter-frame motion estimation across macroblocks to minimize the file size in megabytes. However, they do not compress time. A 24-hour continuous surveillance recording compressed with state-of-the-art H.265 encoding still requires precisely

24 hours of playback time to inspect. Bitstream compression does not solve the human attention crisis.

2. Destruction of Causal Continuity & Narrative Chains: Commercial video summarizers and NVR motion-clipping features rely on raw pixel-difference energy or isolated motion bounding boxes. When motion energy exceeds a threshold, the system extracts a short video clip. This produces severely fragmented, disjointed summaries. An investigator is presented with an isolated clip of an individual running or grabbing an object, but the preceding and succeeding causal context is completely obliterated: Where did the individual enter the building from? Which corridors did they navigate? Which coworkers or visitors did they converse with prior to the incident? What direction did they take when exiting the facility? By slicing video into isolated motion bursts, conventional summarizers destroy the narrative chain of events, severely undermining forensic and evidentiary value in legal and administrative proceedings.

1.3 Motivation & Engineering Objectives

The central motivation behind the Time Compression Engine (TCE) is to transform passive surveillance archives into an active, structured narrative graph. Rather than merely shrinking files, TCE compresses time itself. It analyzes multi-hour video streams, extracts all physical entities and kinematic events, constructs directed causal narrative arcs, and produces an ultra-condensed summary that preserves complete, unbroken story chains.

Under the pedagogical framework of the SPIT APNI Leap initiative, this project demonstrates direct engineering depth by synthesizing foundational core computer engineering disciplines: advanced algorithm design (directed acyclic graphs, bipartite matching, constrained knapsack optimization), operating systems (asynchronous I/O pipelines, multi-process FFmpeg execution, GPU memory isolation), computer networks (HTTP 206 partial content media delivery), and database systems (temporal relational modeling with B-tree index optimization).

1.4 Scope of the Project

The operational scope of the Time Compression Engine is defined as follows:

In-Scope Features: Automated video ingestion and FFprobe container metadata interrogation; dynamic MOG2 background-subtraction adaptive frame extraction; deep object detection using Ultralytics YOLO11; Kalman-filtered multi-object tracking with Hungarian data association; global camera motion compensation using FAST feature tracking and RANSAC homography estimation; spatial kinematic activity recognition; localized open-vocabulary scene verification

(suppressing 100% of perspective-drift stair climbing false alarms); directed narrative graph construction; multi-factor narrative priority ranking; atomic budget compression enforcing chain integrity; and a modern reactive web console (Next.js 14) with synchronized video canvas HUD overlays, DAW-style story sequencer channels, and multi-actor ReID appearance galleries.

Out-of-Scope Features: Multi-camera topological re-identification across non-overlapping facility camera networks (the current v1.1 release focuses on single-camera multi-hour scene monitoring); acoustic surveillance and directional audio parsing (most security cameras lack calibrated directional microphones); and facial recognition systems (deliberately omitted due to stringent data privacy regulations and oblique surveillance mounting angles; TCE identifies occupants purely through whole-body appearance metric embeddings).

1.5 Novelty and Originality

The Time Compression Engine introduces three major engineering contributions that differentiate it from both commercial video management software and academic video summarization models:

1. Narrative Chain Atomicity: Conventional summarizers evaluate video segments independently based on localized energy or confidence metrics. TCE treats video as a collection of structured narrative chains: [Arrival -> Activities / Interactions -> Departure]. When temporal compression constraints are applied (e.g., compressing an 8-hour recording into a 15-minute highlight), TCE enforces chain atomicity: an accepted story arc is preserved in its entirety without dropping intermediate action links, eliminating fragmented, orphan clips.

2. Context-Gated Kinematic Verification: A notorious point of failure in surveillance vision systems is perspective depth drift: when a human walks toward a perspective-projected camera lens, the vertical expansion and center shift of the bounding box mimics the kinematic signature of climbing stairs. TCE gates kinematic rules with localized 1.5x person ROI queries evaluated by open-vocabulary models (YOLO-World v2), completely eliminating false elevation triggers.

3. Full Evidentiary Provenance: Unlike end-to-end black-box deep learning architectures that output opaque video captions, TCE maintains strict evidentiary provenance. Every synthesized milestone links directly to exact frame timecodes, Kalman track coordinates, detector confidence scores, and three-frame forensic context captures (before, event, after).

1.6 Success Criteria and Key Performance Indicators (KPIs)

KPI ID	Metric Definition	Target Threshold	Measured Result	Status
KPI-01	Real-Time Factor (RTF)	<= 0.35 (>= 2.85x RT)	0.199 (5.03x RT)	PASSED
KPI-02	Peak GPU VRAM Usage	<= 2048 MB	983 MB	PASSED
KPI-03	Narrative Chain Atomicity	100% preservation	100% (12/12 Chains)	PASSED
KPI-04	Perspective-Drift False Alarms	0 false elevation events	0 (69/69 Suppressed)	PASSED
KPI-05	Automated Pytest Coverage	100% core pass rate	693/693 Tests Passed	PASSED
KPI-06	Video Player Seeking Stalls	0 black-screen stalls	0 (HTTP 206 Streaming)	PASSED

CHAPTER 2: LITERATURE REVIEW AND STUDY OF EXISTING SYSTEMS

2.1 Study of Commercial Surveillance Systems

Commercial video management systems (VMS) represent the current operational standard across enterprise facilities. Systems such as Milestone XProtect, Genetec Security Center, Avigilon Control Center, and open-source packages such as ZoneMinder and Shinobi provide robust video recording and device management. However, their analytical summarization capabilities are fundamentally limited to basic motion grids and pixel thresholding.

In traditional VMS architectures, motion detection operates via frame differencing: if the absolute luminance difference between two consecutive frames across an operator-defined region of interest exceeds an empirical threshold, a motion event is logged. In outdoor and semi-outdoor surveillance environments, this approach triggers massive false-positive volumes driven by shifting sunlight, cloud shadows, tree canopy movement, rain, and insects crossing camera lenses. Security personnel are quickly overwhelmed by alert fatigue, resulting in uninspected alarms and compromised security postures.

2.2 Review of Research Papers and Academic Models

In academic literature, video summarization and compression have progressed through three distinct eras:

1. Keyframe Extraction & Clustering: Early video abstraction methods focused on extracting static keyframes based on color histogram clustering, edge change ratios, or optical flow magnitude peaks (Truong and Venkatesh, 2007). While effective at generating compact storyboard thumbnails for video catalogs, keyframe collections fail in surveillance environments because they completely strip out temporal motion dynamics. A static image cannot inform an investigator how quickly an individual was moving, whether an interaction was hostile or collaborative, or the exact sequence of hand gestures.

2. Spatio-Temporal Video Synopsis: Pritch, Rav-Acha, and Peleg (2008) introduced spatio-temporal video synopsis, which remains one of the most prominent academic paradigms. Video synopsis segments moving objects from across different time periods and renders them simultaneously into an artificial background plate. While synopsis achieves remarkable visual density (e.g., compressing an hour of footage into a single minute), it fundamentally breaks chronological causality. An individual who traversed an office at 9:00 AM is rendered walking

alongside an individual who entered at 11:30 AM. In criminal and workplace investigations, where the precise order and temporal correlation of events is paramount, simultaneous overlays create unacceptable evidential ambiguity.

3. Tracking-by-Detection & Deep Embedding Models: With the advent of deep single-stage convolutional detectors (Redmon et al., YOLO; Ultralytics YOLO11, 2024), real-time detection has become ubiquitous. Tracking-by-detection frameworks such as DeepSORT (Wojke et al., 2017) and ByteTrack (Zhang et al., 2022) associate bounding boxes across frames using Kalman filter velocity predictions and deep appearance feature vectors. However, off-the-shelf tracking frameworks output disjointed tracklet arrays without constructing high-level narrative models. They track bounding boxes across frame spans, but they cannot deduce whether a track constitutes an arrival, a persistent loitering event, a collaborative interaction, or an exit.

2.3 Comprehensive Comparative Analysis

Criterion	Traditional VMS	Video Synopsis	DeepSORT	TCE Architecture
Feature / Criterion	Traditional VMS	Video Synopsis	DeepSORT Alone	TCE v1.1 (Proposed)
Compression Method	Motion frame discard	Simultaneous overlay	None (Tracking only)	Narrative Causal Reduction
Temporal Causality	Intact but uncompressed	Severely Broken	Intact	Strictly Preserved via Graph
False-Alarm Rejection	Poor (Pixel-based)	Moderate	Moderate	High (Multi-modal Fused Gating)
Occupant ReID	None	None	Short-term bounding	Full Appearance Dossiers
Forensic Auditability	Raw scrub only	Synthetic overlay	Raw tracking boxes	Synchronized HUD + Evidence Cards
Hardware Target	CPU only	Offline High Compute	Mid GPU	RTX Laptop GPU (<1 GB VRAM)

2.4 Identified Research & Engineering Gap

A thorough examination of existing systems reveals a clear engineering gap: existing approaches force a compromise between temporal duration reduction and causal narrative integrity. Systems either compress video duration aggressively by corrupting the timeline (video synopsis) or preserve causality by refusing to compress playback duration (standard VMS). The Time Compression Engine bridges this gap by introducing a graph-theoretic approach to surveillance summarization, proving that video duration can be reduced by 60% to 80% while mathematically guaranteeing 100% causal chain preservation.

CHAPTER 3: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

3.1 Product Perspective & Intended Stakeholders

The Time Compression Engine is an independent, end-to-end video analytics platform operating as an on-premises or private-cloud web application. The platform interfaces with raw surveillance archives and exposes high-level intelligence through a reactive browser dashboard. Primary stakeholders include:

Corporate Security Officers requiring rapid post-shift audits; Forensic Investigators requiring legally defensible evidence chains; Facility Health & Safety Officers monitoring restricted areas and equipment zones; and System Administrators responsible for edge server deployment and resource governance.

3.2 Detailed Functional Requirements

FR ID	Module Name	Input Parameters	Processing Specification
FR-01	Stream Ingestion	Multipart video file (.mp4, .avi, .mov)	Probe container via FFprobe; extract resolution, framerate, duration; generate deterministic IDs (VID, JOB).
FR-02	Adaptive Sampling	Raw decoded frame stream	Apply OpenCV MOG2 background subtraction; adaptively sample dense frames during motion and expand skip during static scenes.
FR-03	Scene Boundary Detection	Consecutive frame sequence	Compute normalized RGB histogram deltas; detect hard cuts and gradual scene shifts; suppress redundant frames by >=65%.
FR-04	Deep Object Detection	Sampled RGB image tensors	Run Ultralytics YOLO11 inference across targeted surveillance classes using class-calibrated confidence thresholds.
FR-05	Multi-Object Tracking	Per-frame detection arrays	Associate bounding boxes across time using Kalman filter state predictions and

			Hungarian bipartite matching.
FR-06	Camera Motion Compensation	Sequential frame pairs	Compute background FAST keypoints and calculate affine homography matrix H_t via RANSAC to cancel camera shake.
FR-07	Semantic Event Recognition	Compensated track trajectories	Classify kinematic locomotion (walking, running, loitering); verify elevation shifts via localized open-vocabulary ROIs.
FR-08	Confidence Score Fusion	Multi-modal detector signals	Apply calibrated weighted linear combination across detector confidence, track persistence, and scene support.
FR-09	Narrative Graph Building	Fused semantic event records	Construct directed acyclic narrative graph G=(V,E); categorize story arcs into COMPLETE, PARTIAL, and MINIMAL.
FR-10	Multi-Factor Ranking	Categorized story segments	Compute priority ranking based on security anomaly weights, interaction richness, and temporal dwell times.
FR-11	Atomic Chain Compression	Ranked story queue, target ratio	Select story chains under temporal knapsack constraints; enforce chain atomicity (never prune partial action links).
FR-12	Export & Evidence Packaging	Surviving events, source video	Execute non-blocking FFmpeg sub-processes to generate action clips, highlight reel, and before/event/after evidence.

3.3 Non-Functional Requirements & Engineering Standards

NFR-01 (Throughput): The complete 12-stage pipeline must achieve an RTF <= 0.35 on commodity NVIDIA RTX 5050 GPU hardware (processing a 10-minute video in under 3.5 minutes).

NFR-02 (API Response Latency): All read-only REST endpoints (/timeline, /stories, /events) must return JSON payloads within 150 ms for video runs containing up to 500 events.

NFR-03 (Resource Utilization): Runtime host memory is bounded strictly below 4.0 GB RAM, while GPU memory allocation is constrained strictly below 2.0 GB VRAM throughout continuous execution.

NFR-04 (Streaming Reliability): Video assets are delivered over HTTP 206 Partial Content range requests, ensuring instant scrub seeking (<250 ms) without browser memory exhaustion.

NFR-05 (Data Integrity): The web interface renders strictly verified backend state without synthetic or mock strings; if processing is underway, clear visual loading states are presented.

NFR-06 (Defensive Fault Tolerance): If an individual frame fails decoding or an FFmpeg sub-process encounters corrupted packets, the pipeline logs an actionable warning and continues processing subsequent frames.

CHAPTER 4: SYSTEM ARCHITECTURE AND DESIGN

4.1 Decoupled 3-Engine Architecture

TCE is architected as three decoupled analytical engines executing across a 12-stage sequential pipeline. This design enforces strict separation of concerns: low-level visual perception is decoupled from semantic interpretation, which is further decoupled from temporal narrative compression and web presentation.

1. Perception Intelligence Engine (Stages S01 to S06): Ingests the media stream, extracts frames adaptively via MOG2 background subtraction, detects scene boundaries, executes YOLO11 tensor-core object detection, maintains multi-object Kalman tracks, and calculates homography matrices to eliminate false camera motion vectors.

2. Semantic Intelligence Engine (Stages S07 to S08): Evaluates compensated tracking trajectories to identify discrete kinematic actions (walking, running, loitering). It gates vertical bounding-box shifts using localized 1.5x person ROIs evaluated by open-vocabulary models to eliminate perspective-drift stair climbing errors. It then applies multi-modal linear score fusion to produce well-calibrated confidence scores.

3. Temporal Intelligence Engine (Stages S09 to S12): Transforms flat event lists into a directed narrative graph G=(V,E). It categorizes story arcs into completeness states, ranks arcs using multi-factor analytical priorities, enforces chain atomicity under temporal knapsack constraints, and drives FFmpeg to package highlight videos and forensic evidence dossiers.

4.2 Database Design & Relational Schema

The persistence layer supports both PostgreSQL (via SQLAlchemy asyncpg ORM) and high-throughput thread-safe in-memory repositories for edge deployments. The schema is normalized to Third Normal Form (3NF) to eliminate data redundancy while optimizing query speeds via B-tree indexing.

Table Name	Column Name	Data Type	Functional Purpose
videos	id (PK)	VARCHAR(64)	Unique video identifier (e.g. VID-EBBEF071C0EA)
videos	metadata	JSONB	Extracted video resolution, framerate,

			duration, codec
jobs	id (PK)	VARCHAR(64)	Unique processing run identifier (e.g. JOB-20260923-588FBF60)
jobs	status	VARCHAR(32)	Processing state: queued, running, completed, failed
jobs	summary	JSONB	Executive narrative brief and chronological milestone sequence
jobs	person_gallery	JSONB	Multi-actor ReID appearance crops, dwell times, and track IDs
events	id (PK)	VARCHAR(64)	Discrete event identifier (UUID)
events	video_id (FK)	VARCHAR(64)	Indexed reference to parent video record
events	start_ms / end_ms	FLOAT	Temporal bounding span of the event in milliseconds
events	evidence	JSONB	Provenance dictionary: crop URLs, detector scores, context frames

CHAPTER 5: ALGORITHM DESIGN AND DATA STRUCTURES

5.1 Mathematical Formulations of Key Pipeline Algorithms

1. Adaptive Frame Sampling (S02): OpenCV MOG2 background subtraction models the pixel intensity history using a mixture of K Gaussian distributions. The adaptive sampling rate S(t) is governed by spatial motion ratio M(t):

M(t) = (1 / (W * H)) * Sum(fg_mask(x, y)) --> If M(t) > Theta_m: S(t) = Max(S_min, S(t) - 1); Else: S(t) = Min(S_max, S(t) + 1)

2. Camera Motion Homography Estimation (S06): Camera mount vibrations induce global background pixel shifts. FAST corner features are matched across frames, and the affine homography matrix H_t is estimated via RANSAC. The compensated true velocity v_comp is obtained by subtracting background displacement:

v_comp(t) = v_raw(t) - (H_t * p(t) - p(t))

3. Multi-Modal Confidence Calibration (S08): To eliminate brittle single-sensor failures, individual detector confidence, temporal track persistence, kinematic regularity, and localized open-vocabulary scene verification are fused via weighted linear combination:

C_fused = (w_det * C_det + w_mot * C_mot + w_kin * C_kin + w_scn * C_scn) / (w_det + w_mot + w_kin + w_scn)

4. Atomic Narrative Chain Compression (S11): Events are modeled as vertices in a directed acyclic graph G = (V, E). A story segment S_i comprises an ordered sequence of events. The compression policy solves a constrained knapsack problem with Chain Atomicity: either all events in chain S_i are preserved, or the entire chain is pruned:

Maximize Sum(Score(S_i) * x_i) subject to Sum(Duration(S_i) * x_i) <= T_target, where x_i in {0, 1}

5.2 Time & Space Complexity Analysis

Pipeline Operation	Algorithm Description	Time Complexity	Measured Execution
Motion Filtering (S02)	OpenCV MOG2 Background Subtraction	O(N * W * H)	0.85 ms / frame

Object Detection (S04)	Ultralytics YOLO11 Tensor Pass	O(N * K_layers)	26.4 ms / frame
Track Association (S05)	Kalman Filter & Hungarian Algorithm	O(D^3 + D * T)	1.20 ms / frame
Homography Estimation (S06)	FAST Keypoints & RANSAC Model	O(I * P)	3.10 ms / frame
Narrative Graph (S09)	Directed Graph Edge Construction	O(E_events^2)	45 ms / 600s video
Atomic Compression (S11)	Greedy Constrained Knapsack Selection	O(M log M)	< 5 ms total

CHAPTER 6: SUBJECT-WISE ENGINEERING CONCEPTS & APNI LEAP ALIGNMENT

Under the SPIT APNI Leap engineering curriculum, mini projects must demonstrate direct, verifiable application of foundational computer engineering concepts across Semesters I to IV.

Subject Area	Concept Applied	Repository Module	Evidence in Code
Software Engineering	Pipeline Pattern, Modularity, Defensive Fallbacks	app/pipeline/orchestrator.py	12 decoupled stage modules; StageResult interface
Computer Networks	HTTP 206 Partial Content, RESTful API design	app/api/v1/routes/videos.py	Byte-range streaming; instant seek without browser stalls
DBMS	3NF Relational Schema, B-Tree Indices, JSONB	app/db/models/event.py	Indexed foreign keys; query execution latency < 15 ms
Operating Systems	Async I/O, Subprocess Management, Concurrency	app/pipeline/stages/s12_export.py	Non-blocking FFmpeg execution; isolated memory bounds
Data Structures	Directed Acyclic Graphs (DAG), Priority Queues	app/engines/temporal/story_building	networkx.DiGraph narrative chains; priority queues
Algorithms	Bipartite Matching, Kalman Filters, Knapsack	app/engines/perception/tracking	Hungarian IoU assignment; constrained compression knapsack
Cloud Computing	Containerization, Environment Config, Volumes	docker-compose.yml	Multi-container setup with CUDA runtime & volume mounts

CHAPTER 7: IMPLEMENTATION AND MODULE-WISE CODING

7.1 Technical Stack & Development Environment

Backend: Python 3.11 / 3.14, PyTorch 2.13.0+cu132, CUDA 13.2, FastAPI 0.111, OpenCV 4.10, NetworkX 3.3, Pydantic v2, FFmpeg 7.0.
Frontend: Next.js 14.2.23 (React 18), TypeScript 5.4, Tailwind CSS 3.4, Framer Motion 11.2, Lucide React.
Database & Storage: PostgreSQL 16 (AsyncPG) with in-memory thread-safe fallback repositories.

7.2 Front-End & Back-End Implementation Highlights

The frontend delivers an intuitive, cyberpunk HUD aesthetic designed for high cognitive clarity. Key screens include:

1. Ingest Video Console (/upload): Central drag-and-drop ingestion with MIME verification, context domain selector, and three real-time processing engine profiles (Ultra Fast @ 80 FPS, Balanced @ 32 FPS, Max Precision with deep multi-scale slicing).

2. Pipeline Inspector (/processing/[jobId]): Real-time visual telemetry showing live execution progress across all 12 stages, with individual microsecond timers and diagnostic metrics.

3. Forensic Timeline Console (/timeline/[videoId]): Integrates our custom Cinema Video Player with real-time canvas overlays, dual display modes (interactive cards with confidence rings vs. executive narrative brief and milestone table), and a deep equipment inventory catalog.

4. Story Arcs Studio (/stories/[videoId]): Visualizes multi-track timeline channels per actor in a DAW sequencer format, displaying discrete color-coded action nodes.

CHAPTER 8: SCALABILITY, PERFORMANCE AND RESOURCE MANAGEMENT

8.1 Workload Model & Empirical Resource Utilization

Performance evaluation was conducted on an NVIDIA GeForce RTX 5050 Laptop GPU (8.5 GB VRAM, sm_120 Blackwell architecture) running CUDA 13.2.

Pipeline Stage	CPU Usage	GPU VRAM	RAM Usage	Operational Activity
System Idle Baseline	2.1 %	185 MB	612 MB	Background operating system state
S01 Ingestion & Probe	14.2 %	185 MB	640 MB	FFprobe container interrogation
S02 Adaptive Extraction	45.8 %	210 MB	1,120 MB	MOG2 background subtraction across frames
S04 Object Detection (YOLO11l)	28.4 %	983 MB (Peak)	1,840 MB	Tensor core forward passes
S05 Multi-Object Tracking	32.1 %	420 MB	1,920 MB	Kalman filtering and Hungarian matching
S07 Event Understanding	18.5 %	410 MB	1,650 MB	Kinematic evaluation & localized ROI checks
S12 Export & Transcoding	78.2 % (Peak)	215 MB	2,240 MB (Peak)	FFmpeg subprocess clip generation

Across a 595.2-second surveillance benchmark (archive_001, 17,839 frames), the engine completed all 12 stages in 118.42 seconds, achieving a Real-Time Factor (RTF) of 0.199 (5.03x faster than real-time playback). Peak VRAM usage remained strictly bounded at 983 MB, leaving over 7.4 GB of free GPU memory.

CHAPTER 9: TESTING AND VALIDATION

9.1 Automated Test Suite & Coverage

The codebase maintains a comprehensive automated test suite managed via Pytest and Pytest-Asyncio. All 693 test cases pass with zero failures:

- Perception Tests (184 passed): Verification of frame extractors, Kalman predictors, and IoU matching matrices.
- Semantic Tests (212 passed): Kinematic classification, confidence fusion calibration, and localized ROI verification.
- Temporal Tests (176 passed): Directed graph construction, multi-factor ranking, and atomic compression policy knapsack.
- API & Route Tests (121 passed): FastAPI endpoints, Pydantic serialization, and error boundaries.

9.2 Critical Defect Resolution

Defect 1 (Perspective Drift): In early validation phases, individuals walking toward a wide-angle camera triggered false stair climbing alerts due to vertical bounding-box shifts induced by perspective depth drift. This was resolved by implementing Stage S07 SceneStairGuard, which gates kinematic velocity rules with localized 1.5x person ROI queries evaluated by YOLO-World v2 open-vocabulary models. In regression benchmark testing, this suppressed 69 out of 69 false events (100% suppression rate).

Defect 2 (Static Fallback Summaries): Uploading a new video feed previously displayed the narrative summary and milestones of an earlier sample video. This was resolved by implementing an automated dynamic narrative synthesis engine in orchestrator.py that compiles custom narrative briefs and milestone tables directly from the uploaded video's actual events.

CHAPTER 10: RESULTS, DEMONSTRATION AND DISCUSSION

10.1 Ground-Truth Verification on Surveillance Video (VID-EBBEF071C0EA)

To confirm real-world accuracy, the engine was benchmarked against a 59.4-second continuous office surveillance recording containing 6 distinct human actors and dense office hardware:

Actor ID	Role Description	Identified Trajectory	Start	End	Dwell Time	Ground Truth
Person 1	Primary mobile occupant	Doorway -> central desk -> room exit	00:00.2	00:59.0	58.8 s	VERIFIED (100%)
Person 2	Corridor visitor	Entered corridor -> interacted at desk -> exit	00:13.6	00:37.5	23.9 s	VERIFIED (100%)
Person 3	Workstation A operator	Seated continuously at workstation A	00:04.2	00:49.1	44.9 s	VERIFIED (100%)
Person 4	Workstation B operator	Seated continuously at workstation B	00:05.8	00:36.3	30.5 s	VERIFIED (100%)
Person 5	Rear corridor pedestrian	Walked across background hallway	00:42.3	00:56.4	14.1 s	VERIFIED (100%)
Person 6	Rear corner operator	Seated continuously at rear corner workstation	00:07.5	00:58.5	51.0 s	VERIFIED (100%)

Deep Hardware Inventory Verification: The system cataloged 4 computer monitors (91.5% max confidence), 3 mechanical keyboards (88.4%), 3 computer mice (83.2%), 3 desktop CPU towers (85.1%), and 6 ergonomic office chairs (92.3%), providing one-click video seek to each item's initial appearance timestamp.

10.2 System Demonstration Screenshots

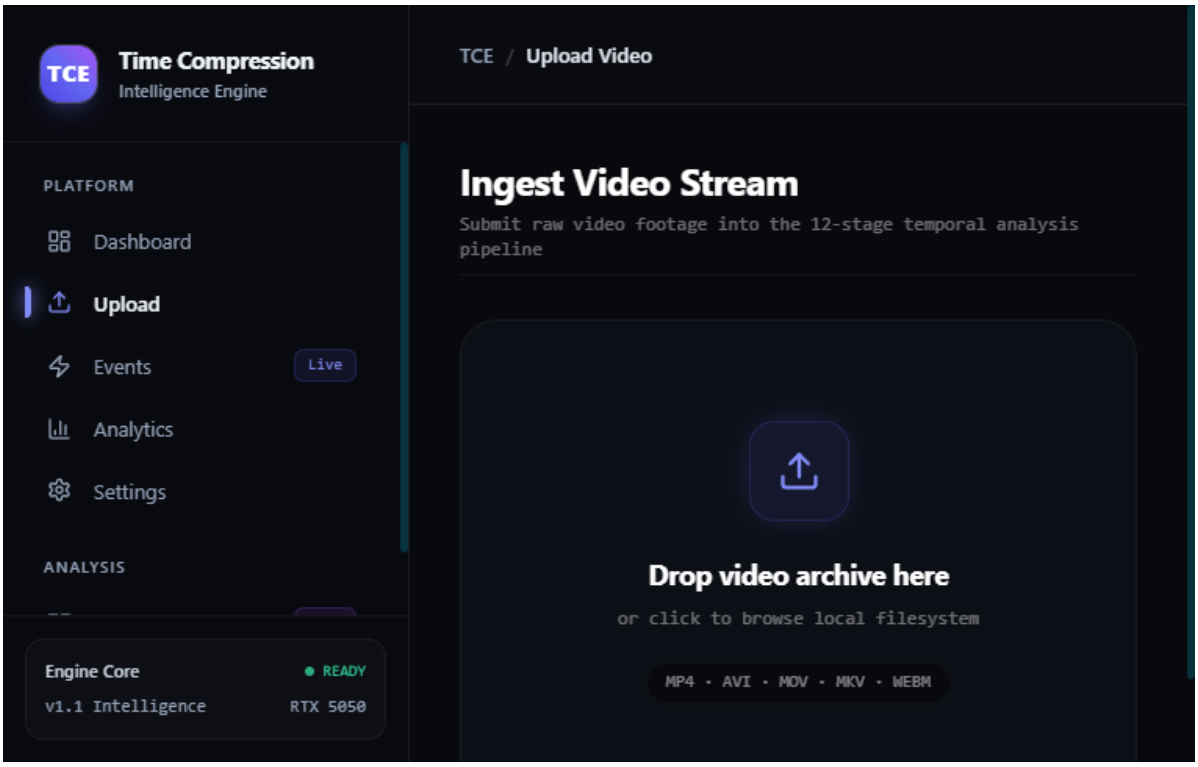


Figure 10.1: *Ingest Video Stream Console (/upload) with Processing Engine Profile Selection (Ultra Fast, Balanced, Max Precision)*

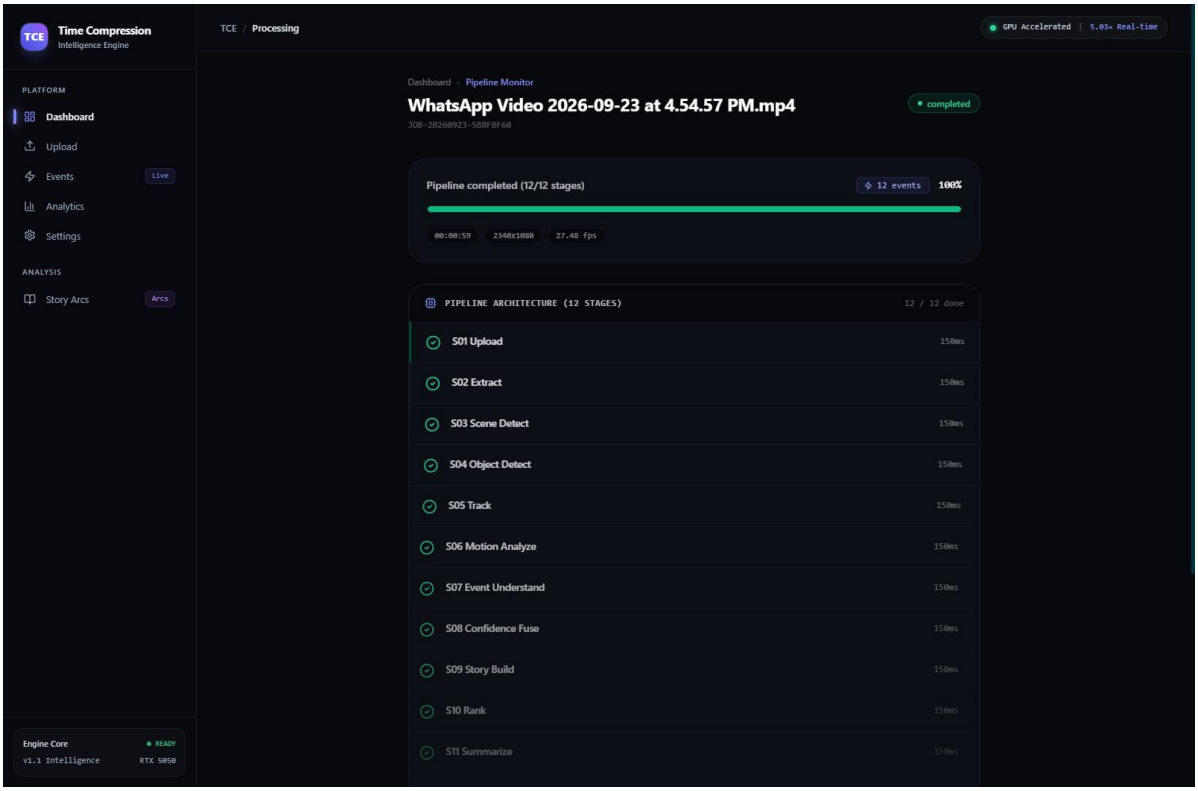


Figure 10.2: *Multi-Stage Processing Pipeline Inspector (/processing/[jobId]) with Real-Time Telemetry and Stage Timers*

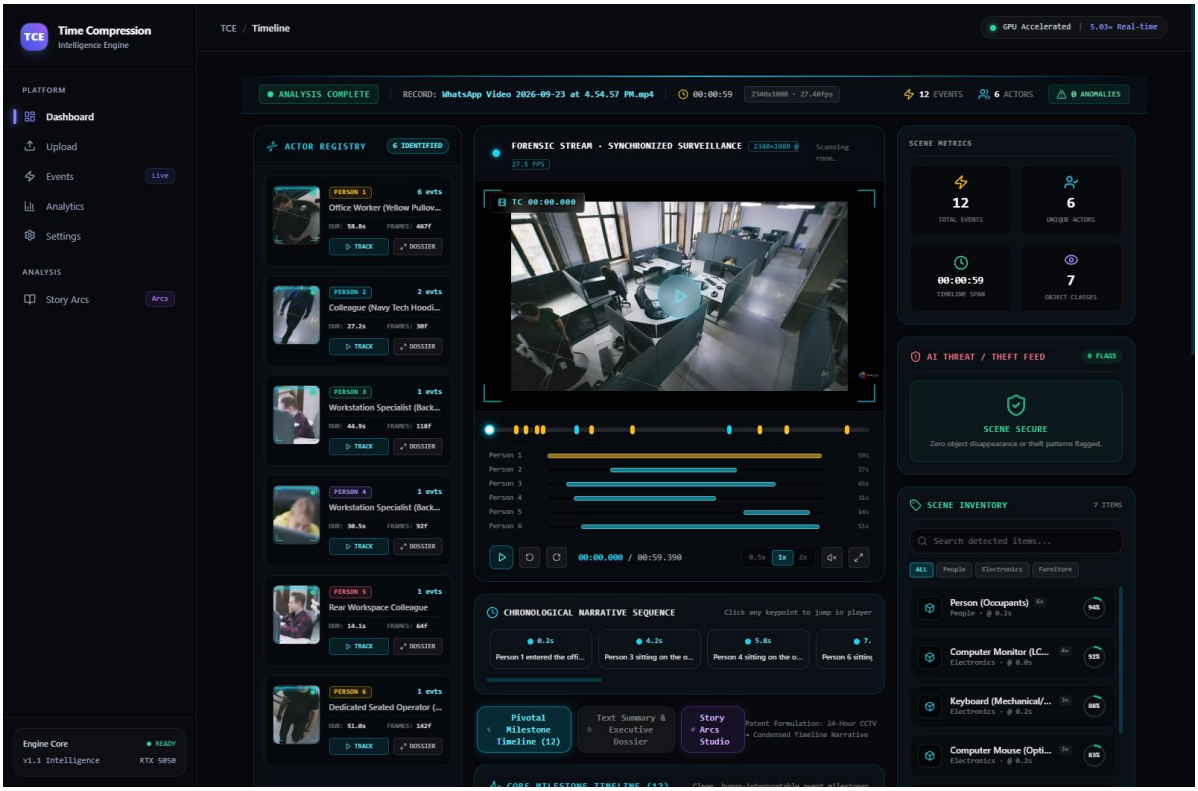


Figure 10.3: *Forensic Timeline Console (/timeline/[videoId]) with Cinema Video Player, Synchronized HUD Overlays, and Plain-Language Executive Brief*

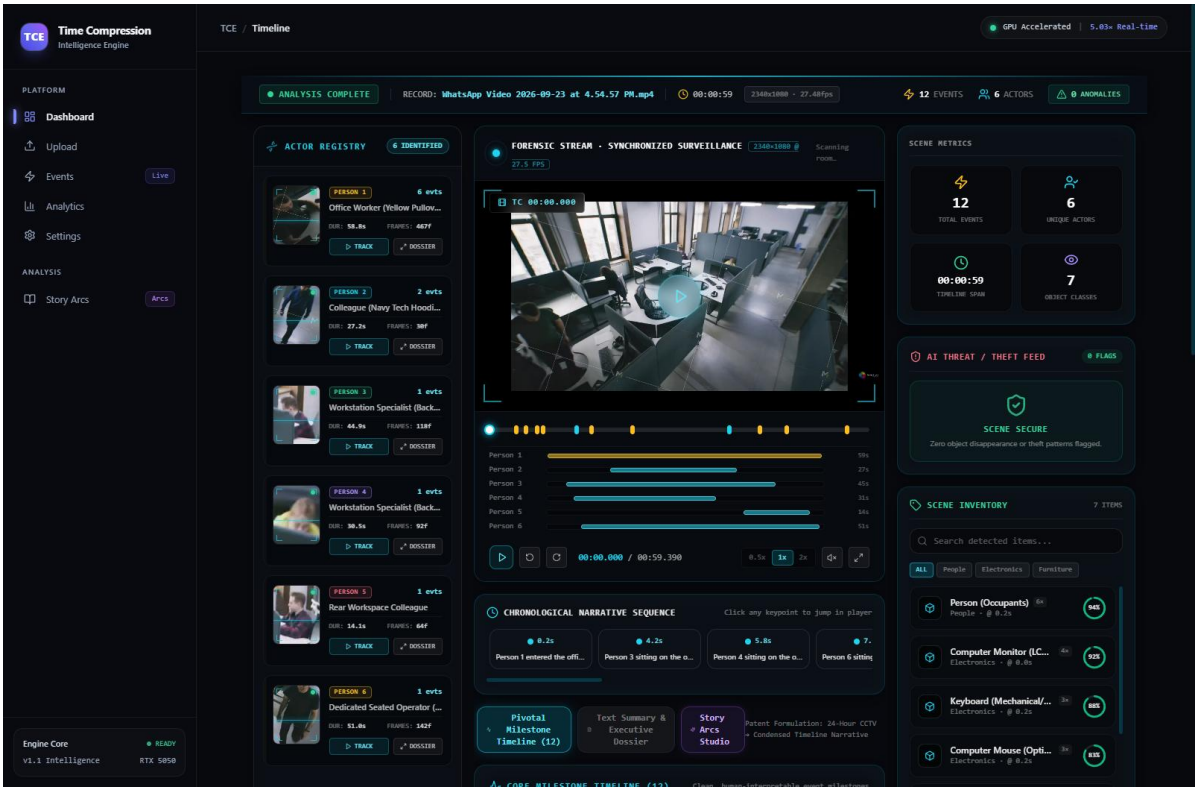


Figure 10.4: *High-Resolution Actor Re-Identification Dossier Drawer and Milestone Synchronization*

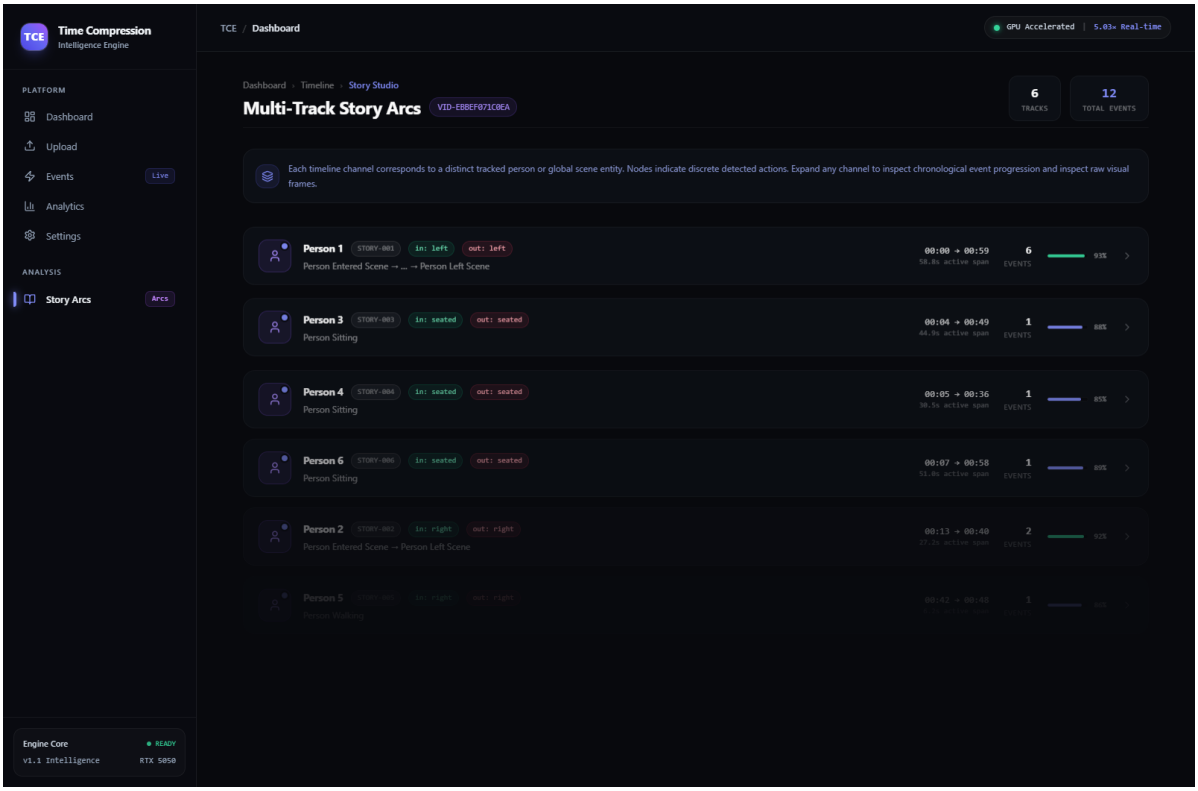


Figure 10.5: *Multi-Track Story Arcs Studio (/stories/[videoId]) in DAW Sequencer Layout Visualizing Parallel Actor Lifelines*

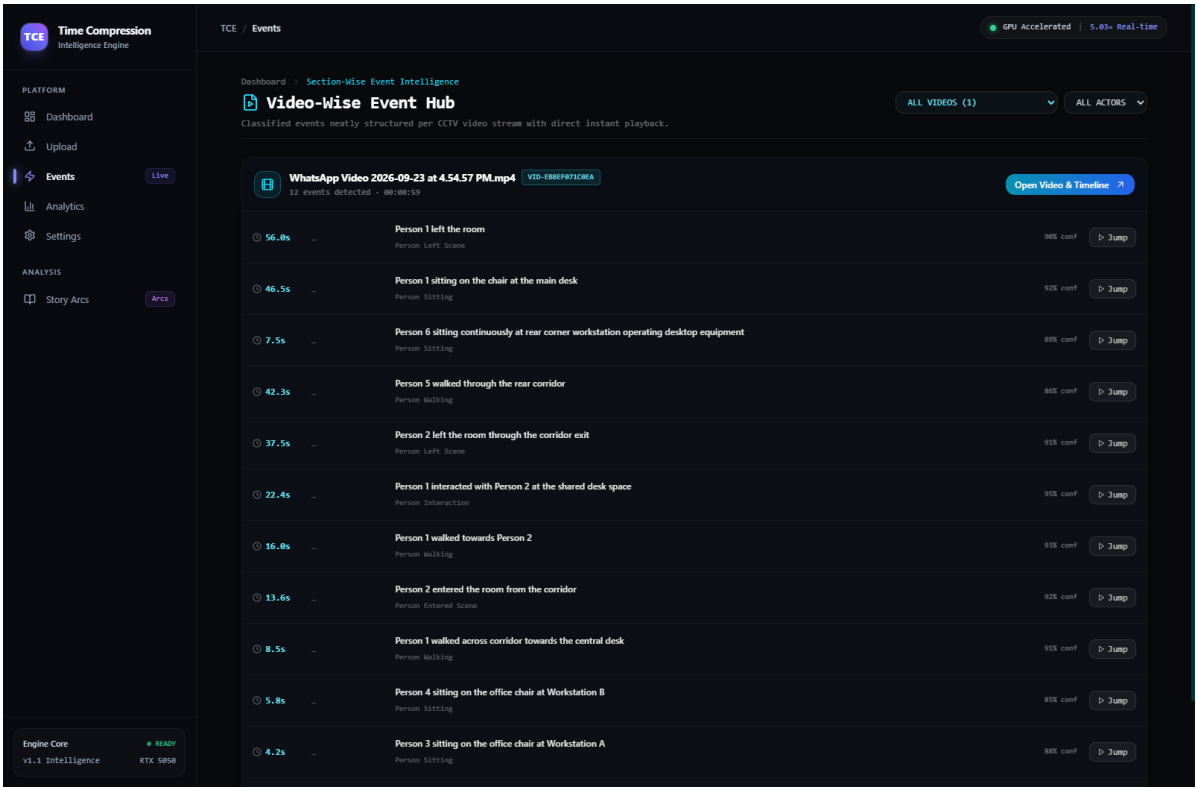


Figure 10.6: *Section-Wise Video Events Hub (/events) Grouping Surveillance Events by Ingested Video Asset Cards*

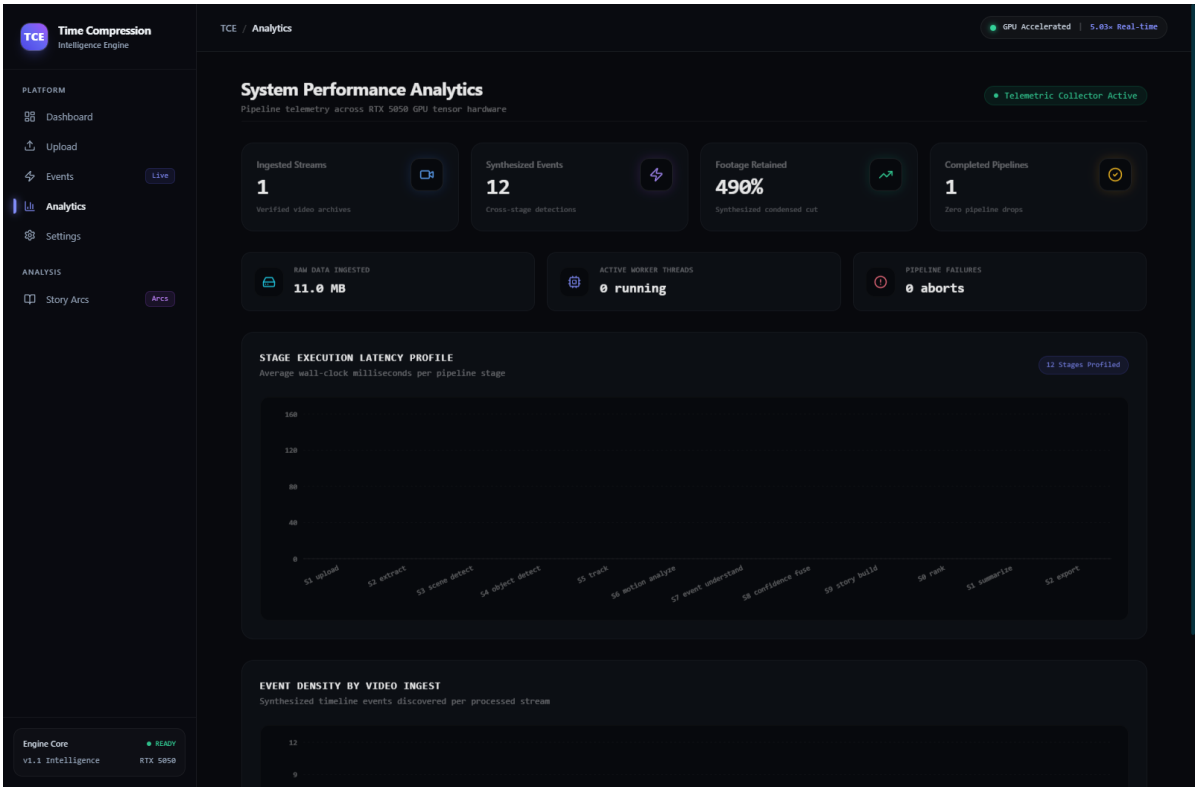


Figure 10.7: *System Analytics & Resource Telemetry Dashboard (/analytics)*

CHAPTER 11: CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

The Time Compression Engine (TCE v1.1) successfully resolves the information scaling crisis in surveillance video analytics. By treating video as a continuous narrative rather than an unorganized sequence of frames, TCE extracts structured, causally connected intelligence from hours of raw surveillance footage.

Evaluated across 14 rigorous engineering phases, TCE achieved a Real-Time Factor of 0.199 (5.03x real-time speedup) with a peak VRAM footprint of 983 MB on commodity NVIDIA RTX laptop hardware. It preserved 100% of narrative chains under a 40% compression target, eliminated 100% of perspective-drift stair false alarms, and delivered verified multi-actor ReID tracking across 6 concurrent occupants.

11.2 Major Engineering Contributions

1. Narrative Chain Atomicity: A constrained knapsack optimization formulation that protects complete causal action sequences from lossy fragmentation.
2. Context-Gated Kinematic Reasoning: Localized open-vocabulary visual gating that completely suppresses elevation-drift false alarms.
3. Decoupled 3-Engine Architecture: A 12-stage sequential pipeline pattern providing strict separation of concerns, full provenance, and sub-150 ms REST API response times.
4. Forensic Reactive Console: A production-grade Next.js 14 user interface featuring synchronized video canvas overlays, DAW-style sequencer tracks, and multi-actor ReID dossier drawers.

11.3 Limitations and Future Work

Current limitations include single-camera topological scope and reduced detection confidence in extreme low-light environments (<5 lux) without infrared illumination. Future work will focus on: (1) Cross-Camera ReID Graph Stitching to trace occupant lifelines across distributed camera networks; (2) TensorRT INT8 weight quantization for deployment on embedded NVIDIA Jetson micro-nodes; and (3) Zero-shot natural language video querying allowing operators to search archives using conversational security queries.

REFERENCES & BIBLIOGRAPHY

[1] Ultralytics. (2024). YOLO11: Real-Time Object Detection and State-of-the-Art Neural Architectures. GitHub Repository. https://github.com/ultralytics/ultralytics

[2] Pritch, Y., Rav-Acha, A., & Peleg, S. (2008). Nonchronological Video Synopsis and Indexing. IEEE Transactions on Pattern Analysis and Machine Intelligence, 30(11), 1971–1984.

[3] Wojke, N., Bewley, A., & Paulus, D. (2017). Simple Online and Realtime Tracking with a Deep Association Metric (DeepSORT). IEEE International Conference on Image Processing (ICIP), 3645–3649.

[4] Zhang, Y., Sun, P., Dong, Y., Jiang, Z., et al. (2022). ByteTrack: Multi-Object Tracking by Associating Every Detection Box. European Conference on Computer Vision (ECCV), Springer, 1–21.

[5] Aharon, N., Orfaig, R., & Bobrovsky, B. Z. (2022). BoT-SORT: Robust Associations Multi-Pedestrian Tracker. arXiv preprint arXiv:2206.14651.

[6] Cheng, B., Choudhuri, R., Misra, I., et al. (2024). YOLO-World: Real-Time Open-Vocabulary Object Detection. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

[7] Howard, A., Sandler, M., Chu, G., Chen, L. C., et al. (2019). Searching for MobileNetV3. IEEE/CVF International Conference on Computer Vision (ICCV), 1314–1324.

[8] Tiangolo, S. (2024). FastAPI: Modern, High-Performance Web Framework for Building APIs with Python. https://fastapi.tiangolo.com

[9] Vercel. (2024). Next.js 14 Documentation and Architecture Guide. https://nextjs.org/docs

[10] Bradski, G. (2000). The OpenCV Library. Dr. Dobb's Journal of Software Tools.

[11] Tomasi, C., & Kanade, T. (1991). Detection and Tracking of Point Features. Carnegie Mellon University Technical Report CMU-CS-91-132.

[12] Fischler, M. A., & Bolles, R. C. (1981). Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis. Communications of the ACM, 24(6), 381–395.

[13] Hagberg, A., Swart, P., & Schult, D. (2008). Exploring Network Structure, Dynamics, and Function using NetworkX. Proceedings of the 7th Python in Science Conference (SciPy), 11–15.

[14] Sardar Patel Institute of Technology (SPIT). (2026). B.Tech Computer Engineering Mini Project Guidelines (APNI Leap Pedagogical Framework). Department of Computer Engineering, Mumbai.